Danmarks Tekniske Universitet

Assignment # 2
Wind Turbine Technology and Aerodynamics

Authors

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Contents

1 Introduction

This report presents the results of the electrical feasibility study of multi-MW variable speed offshore wind turbine installation. The installation consists of 15 identical 10MW DTU wind turbines located 40 km from the shore. Turbines operate under Maximum Power Point Tracking (MPPT) control systems optimizing power extraction based on the wind speed. To assess the feasibility and efficiency of the installation, several calculations are carried out.

Firstly, in section 2 the modelling of the turbine's generator is outlined and the results of its analysis are presented. Secondly, section 3 focuses on the modelling of the transformer and cable connection to the grid. In section 4, wind farm efficiency is calculated and the most efficient MPPT scenario is identified. Lastly, reactive power at the primary transformer winding is found such that reactive power at the point of collection is zero in section 5.

2 Wind Turbine Generator

Wind Turbine Generator is modelled as a permanent magnet synchronous generator operated with a three-phase current. It is represented with the equivalent circuit per phase in Figure 1 with generator characteristic parameters summarized in Table 1.

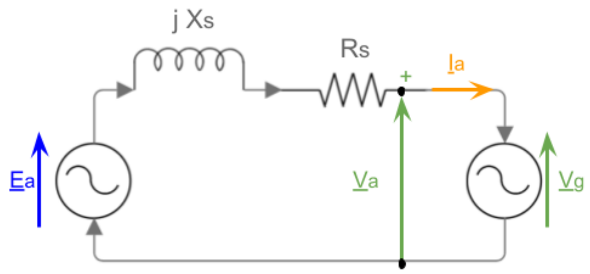

Figure 1: Synchronous generator equivalent circuit

Table 1: Generator characteristic operational parameters

Quantity	Value	Unit
Number of poles N	320	-
Stator inductance L_s	1.8×10^{-3}	H
Stator resistance R_s	94.8×10^{-3}	Ω
Rotor flux linkage φ	19.5	Wb
Nominal apparent power P_{nom}	10.8×10^{6}	W
Nominal mechanical speed ω_{nom}	10	rpm

The equivalent circuit of a synchronous generator represents how it interacts with the external electrical system. The rotor generates an electromotive force (EMF) in the stator by rotating a magnetic field. E_a represents the voltage generated by the flux and is defined as $E_a = \varphi \cdot \omega_e$, where φ is the rotor flux linkage, produced by the field winding from the permanent magnet. Since a real transformer is considered, the experiences an impedance $Z = R_s + X_s i$ which resists the flow of alternating current.

Here some of the power generated is dissipated as heat in the winding, these resistive losses are denoted by R_s and some flux is leaked through the windings and air gap, these are denoted by reactance $X_s = \omega_e \cdot L_s$.

The model consists of several computational steps. First, the relation between mechanical power and the turbine's rotational speed is discussed, followed by calculating the system's electrical angular frequency. Second, the respective procedures for analyzing the

MPPT-Zero Power Angle and MTTP-Zero Beta Angle cases are outlined. Finally, results are presented and discussed.

2.1 Mechanical power and electrical angular frequency

In the system under consideration, wind speed is equivalent to the rotational mechanical speed of the turbine meaning that at 10 m/s a turbine rotates with rotational speed of 10 [rpm]. Moreover, mechanical power P_{mech} is directly proportional to rotational mechanical speed ω as described by the following relation:

$$P_{\text{mech}} = k \cdot \omega^3, \quad \text{where} \quad k = \frac{P_{\text{nom}}}{\omega_{\text{nom}}^3}. \tag{1}$$

The electrical frequency f_e of a synchronous generator is determined as:

$$f_e = \frac{\omega \cdot \frac{N}{2}}{60}[\text{Hz}], \tag{2}$$

where ω stands for a mechanical rotational speed [rpm] and N for the number of generator poles. The electrical frequency f_e can be expressed as angular electrical ω_e with $\omega_e = 2\pi \cdot f_e[\text{rad/s}]$.

2.2 Case 1: MPPT - Zero Power Angle

In the case of Zero Power Angle, the phase current I_a is in phase with terminal voltage V_a. This means that the angle between them is zero as illustrated on the phasor diagram of the equivalent circuit in Figure 2.

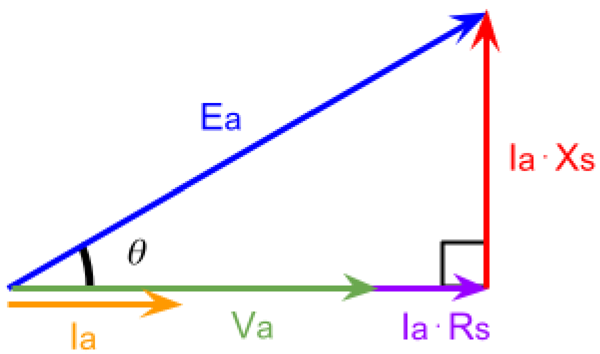

Figure 2: Phasor diagram for MPPT - Zero Power Angle.

The power from the wind turbine's rotors is given in terms of emf E_a and phasor current I_a, $P_{\text{in}} = \text{Real}\,(E_a \cdot I_a^*)$. Since phase current is aligned with terminal voltage (power factor $PF = 1$), the apparent power is only composed of its active (real) power. This allows for obtaining the following relation:

$$P_{\text{mech}} = 3E_a I_a \cos\theta \tag{3}$$

For $PF = 1$, the current is in phase with the voltage, producing a phasor diagram which is a right-angle triangle (Figure 1. This allows to express trigonometric functions of load angle θ in terms of magnitudes of the phasors and using Pythagorean identity as follows:

$$\cos\theta = \frac{P_{\text{mech}}}{3E_a I_a} \qquad\qquad \sin\theta = \frac{X_s I_a}{E_a}$$

$$\begin{aligned} \text{Pythagorean identity} &\Rightarrow \sin^2\theta = (1 - \cos^2\theta) \\ &\Rightarrow \left(\frac{X_s I_a}{E_a}\right)^2 = 1 - \left(\frac{P_{\text{mech}}}{3E_a I_a}\right)^2 \end{aligned} \tag{4}$$

Equation 4 can be rearranged leading to its quadratic form shown in Equation 5, where phase current I_a can be solved analytically. It is important to note that only positive current is considered as the solution in Equation 6 based on the direction shown in the equivalent circuit diagram in Figure 2.

$$(3X_s)^2 \cdot I_a^4 - (3E_a)^2 \cdot I_a^2 + P_{\text{mech}}^2 = 0 \tag{5}$$

$$I_a^2 = \frac{(3E_a)^2 + \sqrt{(3E_a)^4 - 4 \cdot (3X_s)^2 \cdot P_{\text{mech}}^2}}{2 \cdot (3X_s)^2} \tag{6}$$

With I_a, the phase terminal voltage V_a is solved using the Pythagorean theorem as implemented in the following relation:

$$V_a = \sqrt{E_a^2 - (X_s \cdot I_a)^2} - R_s \cdot I_a. \tag{7}$$

2.3 Case 2: MPPT - Zero Beta Angle

In the case of Zero Beta Angle, phase current I_a is aligned with the induced emf E_a as shown on the phasor diagram in Figure 3. The induced emf E_a and inductive impedance X_s are defined identically to case 1.

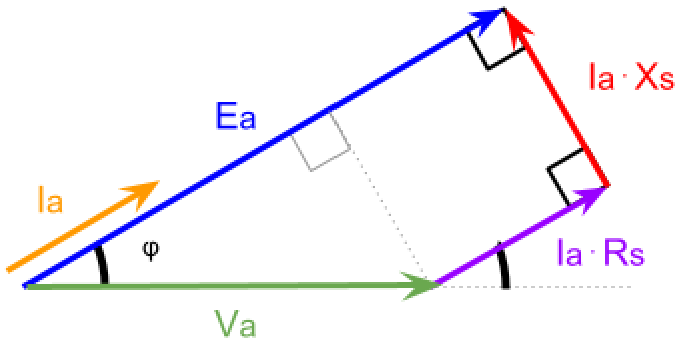

Figure 3: Phasor diagram for MPPT - Zero Beta Angle

The magnitude of the phase current I_a in its phasor form can be directly computed from generator power as it is aligned with the induced emf E_a. This means that the angle θ in equation 3 is zero and yields the equation 8.

$$|\vec{I_a}| = \frac{P_{\text{mech}}}{3 \cdot E_a} \tag{8}$$

Using the magnitude of the phase current, the resistive and reactive voltage drops, $(|I_a| \cdot R_s)$ and $(|I_a| \cdot X_s)$ respectively, are given by Ohm's law. These voltage drops are perpendicular to each other meaning that the Pythagorean theorem can once again be used to find the magnitude of terminal voltage $|V_a|$ as shown in equation 9.

$$|\vec{V_a}| = V_a = \sqrt{(E_a - (|I_a| \cdot R_s))^2 + (|I_a| \cdot X_s)^2} \tag{9}$$

Figure 3, shows that V_a has no imaginary component, meaning the magnitude is equal to the vector $|V_a| = \vec{V_a}$. Given that I_a and V_a are no longer in phase, the angle between the two vectors φ can be found by taking the argument $\tan\varphi = \frac{I_a}{V_a}$. Using this angle the phase current is updated to account for the imaginary component through $\vec{I_a} = |I_a|e^{j\varphi}$.

2.4 Power and efficiency calculations

In both cases, the power and efficiency of the generator are influenced by the losses they experience. Apparent power S of any power source is defined as the product of the voltage of the source with the conjugate of the current flowing through it (Equation 10) leading to a complex number, where P stands for active power and Q for reactive power.

$$S = 3 \cdot V \cdot I^* = P + jQ \tag{10}$$

To calculate the efficiency of the generator η_{gen}, the active power of the circuit at the terminal voltage is divided by mechanical power from the turbine as described by Equation 11.

$$\eta_{gen}\ [\%] = \frac{P}{P_{mech}} \cdot 100 \tag{11}$$

2.5 Results and discussion

Figure 4 illustrates the voltage and current as a function of the mechanical rotational speed, for both cases. The voltage magnitues $|E_a|$ and $|V_a|$ are seen to experience a linear increase with wind speed, whereas current $|I_a|$ increases exponentially. The difference between the two cases is also minimal to none, suggesting that neither case is particularly superior to the other.

Figure 4 also shows the power as a function of rotational speed, which exhibits an exponential increase. This trend is expected because power approximately follows a cubic relationship with rotational speed $P \propto \omega^3$. Here the two cases differ in their reactive power Q. For case 1, where the power factor is equal to one, I_a and V_a are in phase meaning there is no imaginary component in apparent power S. On the other hand, for Beta angle zero in case 2, the angle φ results in an imaginary component of apparent power S. The reactive power for case 2 is negative meaning the current leads the voltage.

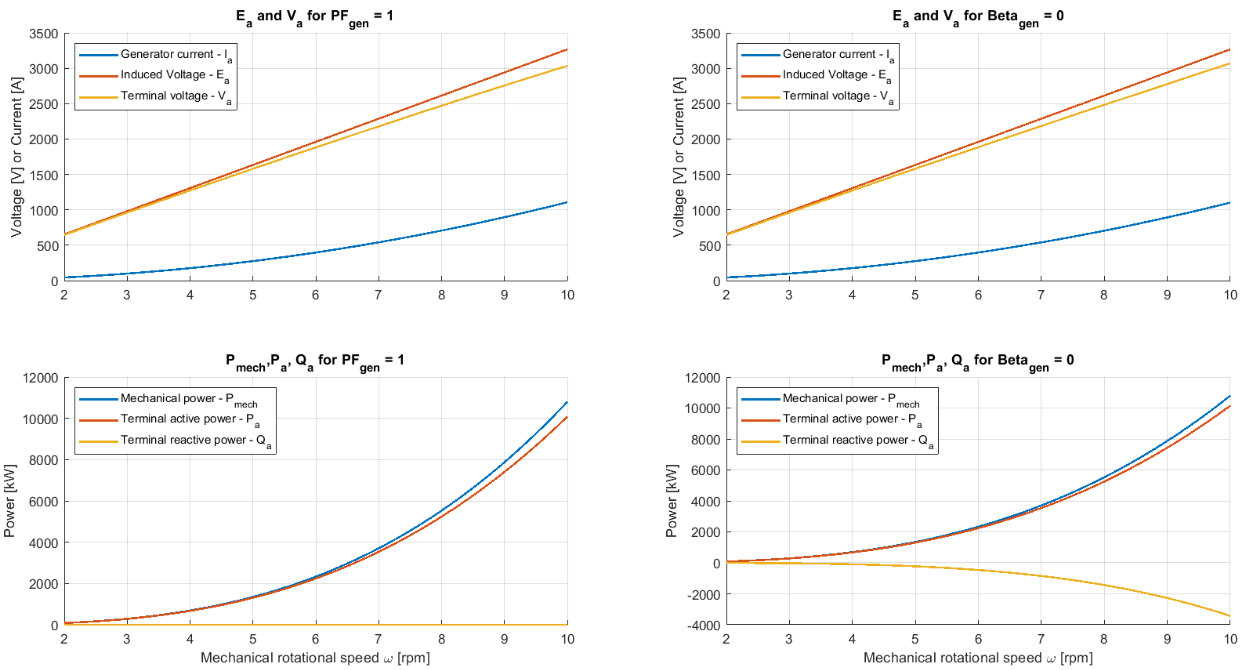

Figure 4: Characteristics generator operational parameters as a function of mechanical rotational speed for two MPPT scenarios.

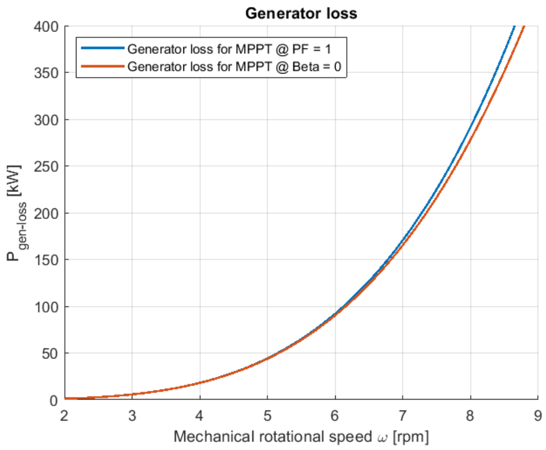

Figure 5: Generator loss as a function of Mechanical rotational speed ω

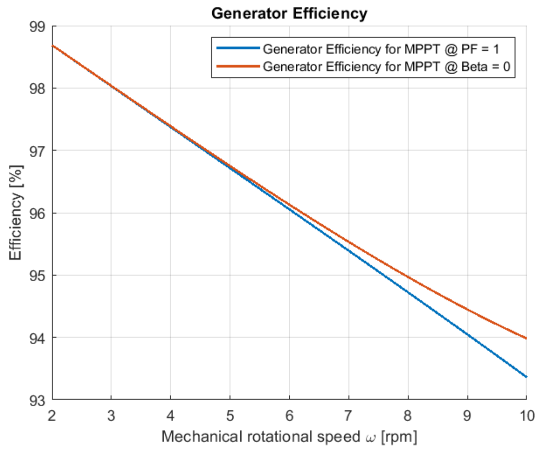

Figure 6: Generator efficiency as a function of Mechanical rotational speed ω

The generator efficiency shown in Figure 6 is found using the equation, 11 and the corresponding power loss shown in Figure 5. From the graphs, it can be concluded that "MPPT - Zero Beta Angle" is a more effective strategy compared to "MPPT - Zero Power Factor" as it leads to lower generator losses. One should note that for low values of mechanical rotational speed both curves align perfectly with each other and the discrepancy starts for values higher than around $\omega = 5$rpm. Nevertheless, the biggest difference in efficiency within the operational range amounts to less than 1% which is considered small.

3 Transformer and Cable connection

The wind turbine transformer is a non-ideal, step-up transformer which, in the discussed case, is characterized by the turn ratio of $a = \frac{4}{33}$. It is modelled by "referring" all electrical quantities to the low voltage side. This results in the transformer equivalent circuit presented in Figure 7. In the figure, impedances representing cable connection to the grid are also included for the sake of completeness.

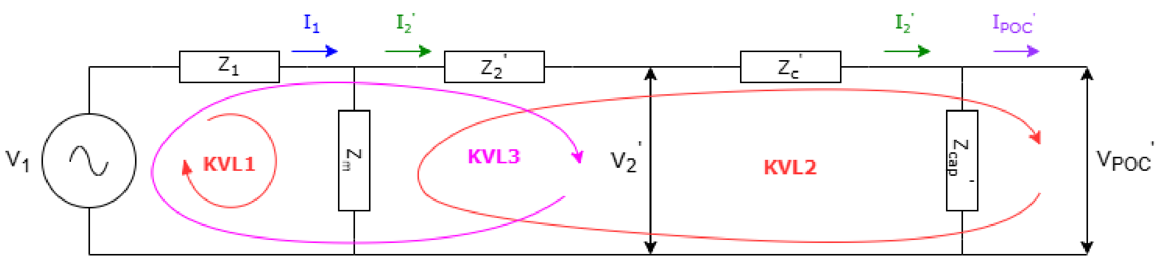

Figure 7: Transformer and grid connection equivalent circuit

All the impedances present in Figure 7 are calculated using equations defined in Table 2 using values from Table 3. It should be noted that, as the transformer is directly connected to the grid, the electrical frequency of the transformer circuit has to match with the electrical frequency of the grid, i.e. $\omega_e = 2\pi f_{grid}$ where $f_{grid} = 50$ [Hz].

Table 2: Impedance Equations for the equivalent transformer circuit.

Quantity	**Equation**
Primary impedance Z_1	$Z_1 = R_1 + j\,\omega_\text{e} L_1$
"Referred to" secondary impedance Z_2'	$Z_2' = R_2' + j\,\omega_\text{e} L_2'$
Magnetizing impedance Z_m	$Z_m = \frac{R_m \cdot j\,\omega_\text{e} L_m}{R_m + j\,\omega_\text{e} L_m}$
"Referred to" capacitor impedance Z_cap'	$Z_\text{cap}' = a^2 \frac{1}{j\,\omega_\text{e} C_\text{cap}}$
"Referred to" cable impedance Z_c'	$Z_\text{c}' = (R_\text{c} + j\,\omega_\text{e} L_\text{c}) \cdot a^2$

Table 3: Circuit parameters for the transformer equivalent circuit.

Quantity	**Value**	**Unit**
Primary inductance L_1	10×10^{-6}	H
Primary resistance R_1	25×10^{-3}	Ω
Core resistance R_m	121.4	Ω
Magnetizing inductance L_m	55×10^{-3}	H
"Referred to" secondary resistance R_2'	25×10^{-3}	Ω
"Referred to" secondary inductance L_2'	10×10^{-6}	H
Cable resistance per length r_c	0.2	Ω/km
Cable inductance per length l_c	0.65×10^{-3}	H/km
Cable capacitance per length c_c	0.05×10^{-6}	F/km
Cable length d	40	km
Line-to-line voltage at POC V_{POC}	33×10^3	V

Having computed all the impedances, Kirchoff's voltage and current laws are used to describe the behaviour of the system. In addition, it is given that from the converter, only the active power P enters the system. These relations are presented in Equation 12.

$$\begin{cases} I_1 = \left(\frac{P}{3V_1}\right)^* \\ -V_1 + I_1 Z_1 + Z_m(I_1 - I_2) = 0 \quad \textbf{(KVL1)} \\ (I_2 - I_1)Z_m + I_2(Z_c' + Z_2') + V_{POC} = 0 \quad \textbf{(KVL2)} \\ I_2 - \frac{V_{POC}}{Z_{cap}'} - I_{POC} = 0 \quad \textbf{(KCL)} \\ V_1 - I_1 Z_1 - Z_2 I_2 - V_2' = 0 \quad \textbf{(KVL3)} \end{cases} \tag{12}$$

Considering the system of equations, there are five unknown quantities, V_1, V_2', I_1, I_2 and I_{POC}. The known quantities consist of the impedances, the active power P from the generator and the magnitude of the voltage at the point of collection $|V_{POC}|$. This magnitude is given by,

$$|V_{POC}| = \frac{33000}{\sqrt{3}} \frac{4000}{33000} \text{ [V]}$$

however it does not mean that V_{POC} consits of only real values. This is a system of equations that must be solved numerically due to the non-linear power term.

3.1 Results and discussion

Consider the results shown in Figure 8.

From the voltage graph, it can be seen that the primary voltage increases to maintain a higher voltage at POC, therefore the step-up transformation is achieved. The behaviour of the active and reactive powers in the transformer is also shown.

4 Wind Farm efficiency

The efficiency of the entire wind farm is defined as the ratio of active power at the point of collection (POC) $Re(S_{\text{POC}})$ to the input mechanical power extracted from the wind by the turbine P_{mech}. The relation is mathematically formulated in Equation 13.

$$\text{Efficiency} \quad \eta = \frac{Re(S_{POC})}{P_{mechanical}} \tag{13}$$

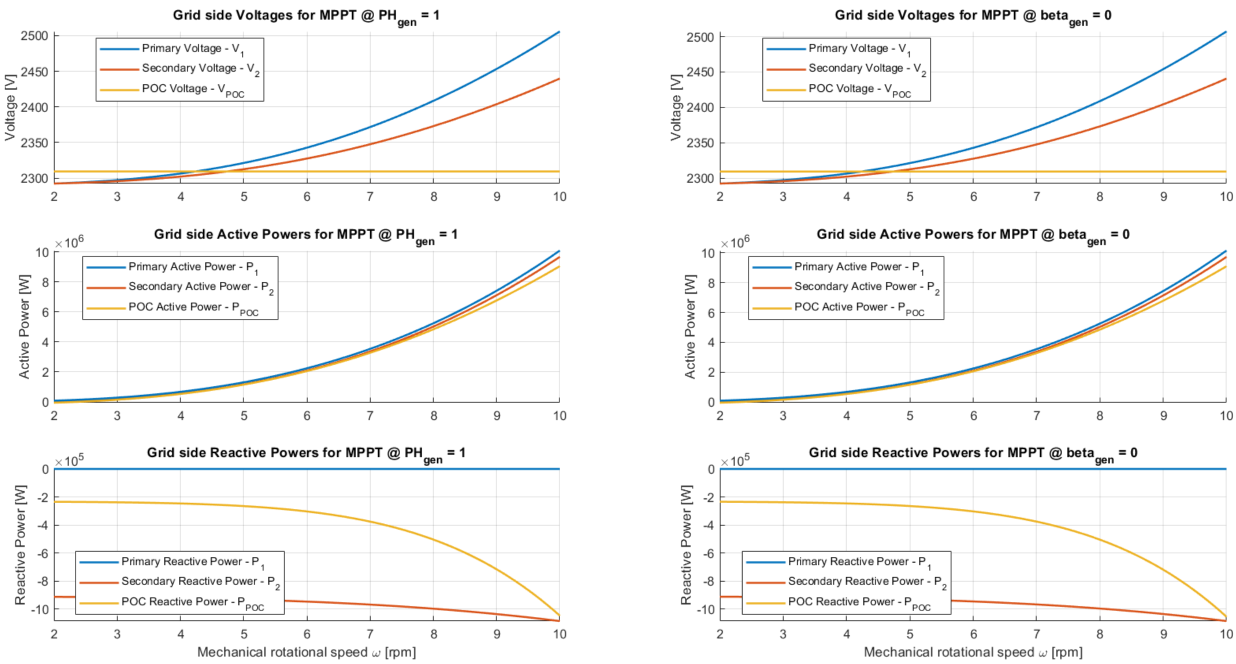

Figure 8: Characteristic transformer operational parameters as a function of mechanical rotational speed for two MPPT scenarios and reactive power forced to be zero at the generator end.

Table 4: Maximum values of efficiency of the wind farm with the corresponding mechanical rotational speeds for each MPPT scenario.

MPPT	Mechanical rotational speed [rpm]	Efficiency [%]
Zero Power Angle	6.7	88.39
Zero Beta Angle	6.8	88.50

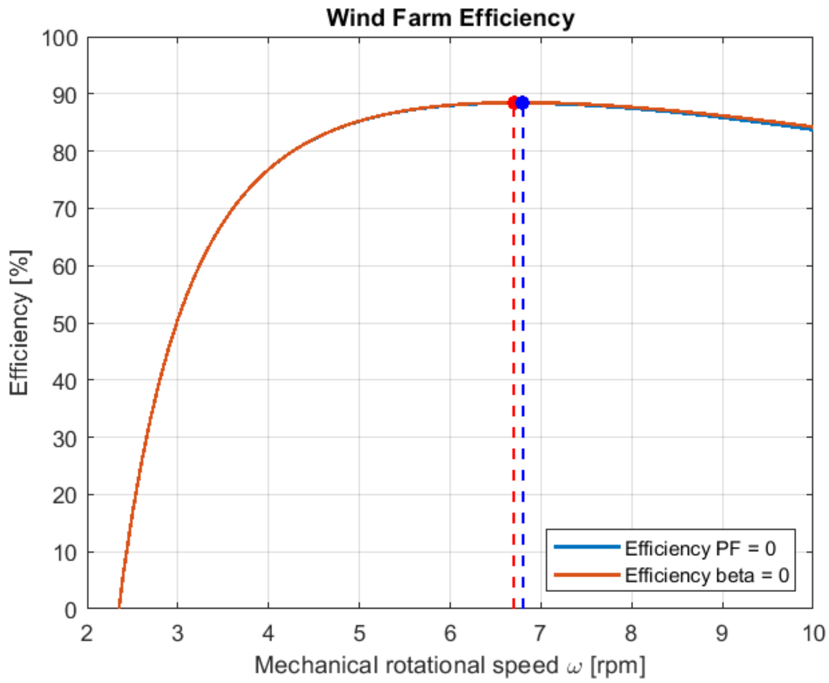

Figure 9: Wind farm efficiency as a function of mechanical rotational speed of the turbine ω for two MPPT scenarios.

Figure 9 presents wind farm efficiency as a function of mechanical rotational speed for two MPPT scenarios. It is visible that both scenarios yield very similar efficiency results. However, it can be concluded that "MPPT - Zero Beta Angle" is more efficient. The highest values of efficiency with the corresponding mechanical rotational speeds they occur at are summarized in Table 4.

5 Desired Reactive Power

To keep the power factor at the power system point of connectivity at unity, meaning $\mathrm{PF_{POC}}$, the current and voltage at the point of collection, I_{POC}, V_{POC} are kept in phase. This means that the angle between them is zero. Using the same approach as described in Section 3, the reactive power term at the transformer input Q_1 is added to equation 12 and solved for. Consider Figure 10 showing the result.

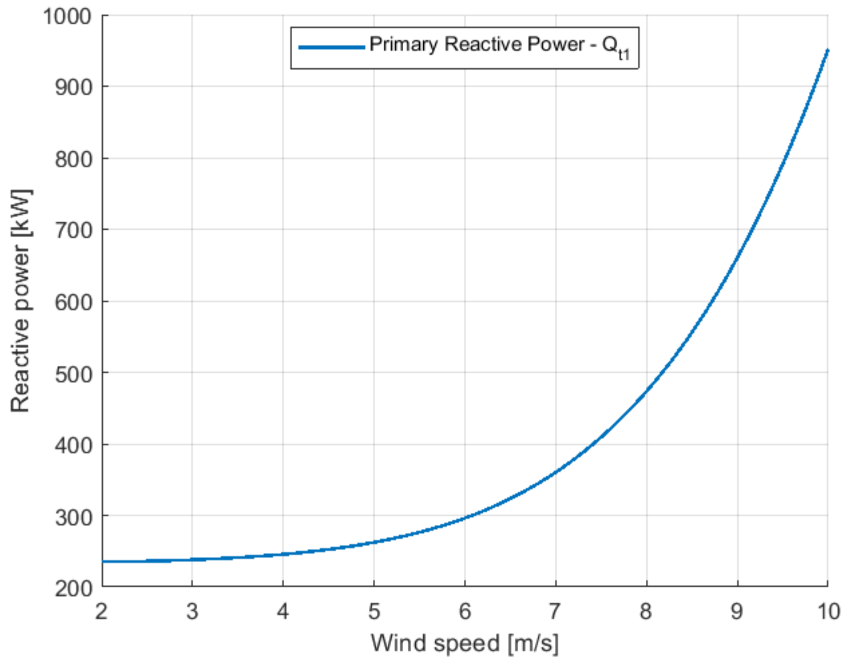

Figure 10: Primary Reactive power required to maintain PF_{POC} at point of collection

The reactive power increases with wind speed so that the power at the collection point is only real. Compared to Figure 8, it can be seen that it is similar to the previously found reactive power POC except positive.

6 Conclusion

To conclude, this paper looks at analyzing the electrical system of a wind turbine to determine its efficiency. Calculations for two different MPPT cases were conducted showing that the Zero Beta Angle suggests slightly better performance. This efficiency however is minimal to none. The reactive power needed from the converter in the system is also found if a Power Factor of one is desired at the point of collection.

List of Figures

List of Tables

References

7 Appendix A

46300 - Group Contract

- **Project Manager:** Jakub Kuleta
- **Main Reporter:** Szczepan Letkiewicz
- **Main Coder:** both

Expectations

We collectively expect an above-average performance. The assignments are not our highest priorities, so a grade of 12 is not likely. Somewhere in the range of 7-10 will be satisfactory for our ambitions.

When

Tasks will be assigned to individual group members collectively, after which each member can complete them at their own time and discretion. An interim deadline will be set to 3 days before the final submission so that each group member can look over the report and make any final adjustments if necessary.

How

The assignment will be divided into sections that require collaboration and which can be completed individually. The fundamentals of BEM will be, for instance, a topic that all members of the group are expected to know on their own, whereas some of the difficult niche questions may be answered by certain members.

Amount

Depending on the amount of work, full flexibility of each member is expected.

Flexibility

All members are expected to complete their tasks and attend meetings arranged beforehand. If any spontaneous meetings were to occur, the other members are to be invited but not required to join.

Presence

Members are expected to be physically present at all pre-arranged meetings. This excludes spontaneous meetings.

Repercussions

A 2-strike system will be enforced: after two strikes, the member will be reported to the Professors.

Signed by - Group 25
12/09/2024

Task division

Task	Jakub Kuleta (%)	Szczepan Letkiewicz (%)
Question 1	50%	50%
Question 2	50%	50%

Table 5: Task Contribution by Group Members